The sharp level-one Fourier weight at density one eighth

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Abstract

The level-one Fourier weight measures the total squared correlation of a Boolean function with the individual coordinates. The proof shows that its maximum at mean 1/8 is 3/64 in every dimension at least three. A codimension-three subcube attains this value. Equivalently, a subset of the Boolean cube with density 1/8 has average Hamming distance at least $(n-3)/2$. The proof establishes a sharp positive-part inequality for Rademacher sums. An even polynomial of degree eighteen bounds the positive part. An exact symmetric moment certificate then bounds the expectation of that polynomial in every dimension. The certificate uses rational coefficients and nonnegative terms. A companion Lean formalization includes the moment identities, the Fourier and distance bounds, and their sharpness witnesses.

1 Result and scope

Let $X = (X_1, \dots, X_n)$ be uniform on $\{-1,1\}^n$, and let $f : \{-1,1\}^n \to \{0,1\}$. Write

$$a = \mathbb{E} f(X), \qquad b_i = \mathbb{E}[X_i f(X)], \qquad W_1(f) = \sum_{i=1}^{n} b_i^2.$$

The following bound holds:

$$\boxed{a = \frac{1}{8} \quad \Longrightarrow \quad W_1(f) \le \frac{3}{64}.} \tag{1}$$

For every $n \ge 3$, equality is attained by the indicator of the subcube $X_1 = X_2 = X_3 = 1$. Consequently the maximum is exactly 3/64 in every admissible dimension, and the dimension-free supremum $W(1/8)$ is 3/64.

In the average-distance formulation, if $A \subseteq \{-1,1\}^n$ and $|A| = 2^{n-3}$, then

$$\boxed{\frac{1}{|A|^2} \sum_{x,y \in A} d_H(x,y) \ge \frac{n-3}{2}.} \tag{2}$$

Equality holds for a codimension-three subcube. Pairs in this formula are ordered, sampled independently, and may coincide.

The result concerns exactly the density 1/8. It does not determine the optimum at every density or the full noise stability of a subcube. No classification of all equality cases is claimed in this note.

2 Relation to previous bounds

The minimum-average-distance problem in Hamming space goes back to Ahlswede and Katona [1]. Fu, Wei, and Yeung obtained linear-programming bounds and solved several cases, including quarter-cube size [2]. Yu and Tan later characterized the asymptotically best bounds from that linear program [3].

Yu explicitly asks for $W(1/8)$ in Section VII, Problem 2, of [4]. Section III-C gives an upper bound of approximately 0.0495142. The codimension-three subcube gives the lower bound $3/64 = 0.046875$. The result here closes this gap by proving the subcube value is an upper bound in every dimension.

The proof combines a polynomial majorant with a dimension-independent moment identity. Its coefficients are rational. Nonnegative Bernstein coefficients establish the one-variable inequalities. A sum of nonnegative symmetric terms establishes the moment inequality. These are established methods. The specific certificate and its sharp bound are the contribution here.

3 The auxiliary sharp Rademacher inequality

For any real coefficients $c_1, \dots, c_n$ with $\sum_i c_i^2 = 1$, put

$$S = \sum_i c_i X_i.$$

The stronger auxiliary estimate is

$$\mathbb{E}(|\sqrt{3}S| - 9/4)_+ \le 3/16, \qquad t_+ = \max(0, t). \tag{3}$$

This constant is attained when $c_1 = c_2 = c_3 = 1/\sqrt{3}$ and all other coefficients vanish: $|\sqrt{3}S|$ is 3 with probability 1/4 and 1 with probability 3/4.

The proof uses a degree-eighteen even polynomial

$$P(t) = \sum_{j=0}^{9} p_j t^{2j} \tag{4}$$

with rational coefficients, given in Appendix A. The two properties required are

$$P(t) \ge \max(0, |t| - 9/4) \quad \text{for all } t \in \mathbb{R}, \tag{5}$$

and

$$\mathbb{E}P(\sqrt{3}S) \le 3/16 \quad \text{whenever } \sum_i c_i^2 = 1. \tag{6}$$

The exact contacts $P(1) = 0$ and $P(3) = 3/4$ explain why equality can occur at the three-coordinate Rademacher sum. Contacts alone do not prove either global inequality; Sections 4–5 provide their certificates.

4 Global verification of the one-variable majorant

Set $p(u) = \sum_{j=0}^{9} p_j u^j$. Exact polynomial division gives

$$p(u) = (u-1)^2 R(u), \qquad \deg R = 7, \tag{7}$$

and

$$P(t) - t + 9/4 = (t-3)^2 T(t), \qquad \deg T = 16. \tag{8}$$

Both R and T are nonnegative on $[0, \infty)$. Here is a finite rational certificate of that assertion, including how it is checked.

For a degree-d polynomial Q on a rational interval $[L, H]$, use the Bernstein representation

$$Q(x) = \sum_{i=0}^{d} \beta_i \binom{d}{i} \left(\frac{x-L}{H-L}\right)^i \left(\frac{H-x}{H-L}\right)^{d-i}. \tag{9}$$

Every basis term is nonnegative on the interval. Thus nonnegative rational β_i prove $Q \geq 0$ there. For the remaining ray, a shifted expansion $Q(B+s) = \sum_i \gamma_i s^i$ with $\gamma_i \geq 0$ proves nonnegativity for $s \geq 0$.

The complete exact R, T, all Bernstein coefficient arrays, and both shifted-tail arrays are in `univariate_certificate.json`. The intervals are:

- R: [0,32], [32,64], [64,80], [80,96], [96,128]; tail begins at 128.
- T: [0,4], [4,5], [5,11/2], [11/2,6], [6,7], [7,8], [8,9], [9,19/2], [19/2,10], [10,12], [12,16]; tail begins at 16.

These are contiguous covers of $[0, 128]$ and $[0, 16]$, respectively. All their Bernstein coefficients and all shifted-tail coefficients are nonnegative rational numbers. Every identity (7)–(9) and every tail identity is explicitly written in the Univariate section of `SharpEighth.lean`, checked by Lean's `ring` tactic, and combined with elementary order reasoning. This proves $R(u) \geq 0$ for $u \geq 0$ and $T(t) \geq 0$ for $t \geq 0$ without approximation.

Equation (7), applied to $u = t^2$, proves $P(t) \geq 0$ for every real t. Equation (8) proves $P(t) \geq t - 9/4$ for $t \geq 0$. Since P is even, apply the latter inequality to $|t|$ to obtain (5).

5 The dimension-independent moment certificate

5.1 Ordered orbit sums

Let $z_i = c_i^2$ and $V = \sum_i z_i$. For a finite list of nonnegative integers $\lambda = (\lambda_1, \ldots, \lambda_\ell)$, define

$$O_\lambda(z) = \sum_{\iota : \{1,\ldots,\ell\} \hookrightarrow \{1,\ldots,n\}} \prod_{r=1}^{\ell} z_{\iota(r)}^{\lambda_r}. \tag{10}$$

The sum is over **ordered injections**. Define $O_{()} = 1$. If $\ell > n$, the sum is empty and equals zero. Permuting the list λ leaves O_λ unchanged. If λ is a partition and $\mathrm{aut}(\lambda)$ is the product of the factorials of its part multiplicities, then the ordinary monomial symmetric polynomial is $m_\lambda = O_\lambda / \mathrm{aut}(\lambda)$.

Two identities control the algebra:

$$V O_\lambda = O_{(\lambda,1)} + \sum_{r=1}^{\ell} O_{\lambda + e_r}, \tag{11}$$

and, when a new coordinate a is adjoined,

$$O_\lambda(a, z) = O_\lambda(z) + \sum_{r=1}^{\ell} a^{\lambda_r} O_{\lambda \setminus r}(z). \tag{12}$$

To prove (11), expand $V O_\lambda$. The coordinate chosen from V either occurs at the unique marked position r in the injection or is new. The first case gives $O_{\lambda+e_r}$, and the second gives $O_{(\lambda,1)}$. To prove (12), classify an injection according to whether it avoids the new coordinate or uses it at its unique position r. Deleting that position is a bijection with injections defining $O_{\lambda\setminus r}(z)$. No division by z_i is used, so zero coordinates and every dimension are covered.

5.2 Rademacher moments

For integers $0 \le j \le 9$, define

$$M_j(z) = \sum_{\lambda \vdash j} \frac{(2j)!}{\mathrm{aut}(\lambda) \prod_r (2\lambda_r)!} O_\lambda(z). \tag{13}$$

Then

$$M_j(c_1^2, \dots, c_n^2) = \mathbb{E} S^{2j}. \tag{14}$$

Indeed, expand S^{2j}. Independence and symmetry kill every monomial containing an odd exponent. The surviving even exponent pattern 2λ has multinomial coefficient $(2j)!/\prod_r (2\lambda_r)!$. Passing from distinct monomials to the ordered-orbit convention divides by $\mathrm{aut}(\lambda)$.

The Lean proof uses a finite recurrence instead of assuming a general multinomial theorem. Averaging the two signs of a new coefficient gives

$$\mathbb{E}(aX_0 + S)^{2j} = \sum_{k=0}^{j} \binom{2j}{2k} a^{2k} \mathbb{E} S^{2(j-k)}. \tag{15}$$

The ten explicit tables $M_0, \dots, M_9$ in the CertificateData section satisfy the identical recurrence under (12); the ten finite coefficient equalities are kernel-checked in the MomentRecurrenceData section. In dimension zero both sides of (14) equal 1 for $j = 0$ and 0 otherwise. Induction on dimension, using (12) and (15), proves (14) for every needed $0 \le j \le 9$. Thus this step is not a check only in a fixed number of variables.

5.3 Weighted squares

For a list λ, real vector $a = (a_1, \dots, a_\ell)$, and scalar b, define

$$G_{\lambda;a,b}(z) = \sum_{\iota \text{ injective}} \left(\prod_r z_{\iota(r)}^{\lambda_r} \right) \left(\sum_r a_r z_{\iota(r)} + bV \right)^2. \tag{16}$$

Both $O_\lambda(z)$ and $G_{\lambda;a,b}(z)$ are nonnegative when all $z_i \ge 0$. Expanding the square and using (11) expresses (16) in the ordered-orbit basis. The coefficients a_r attach to the original positions of λ: the two marked exponents are incremented before sorting the resulting list. This convention matters when equal exponents occur.

The decisive exact identity is

$$\frac{3}{16} V^9 - \sum_{j=0}^{9} p_j 3^j M_j(z) V^{9-j} = \sum_{r=1}^{28} w_r H_r(z), \qquad w_r > 0, \tag{17}$$

where the H_r and weights are given in Appendix B. The first five terms are ordered-orbit monomials of degree nine; the other twenty-three are weighted squares with $|\lambda| = 7$. Thus every term is homogeneous of degree nine.

For clarity, (17) is a finite rational polynomial identity, independent of the numerical values of z or the dimension. Expand (16), repeatedly use (11), collect equal sorted exponent lists, and compare rational coefficients. There are exactly thirty partitions of nine. One coefficient is zero on both sides; the other twenty-nine normalized coefficients agree. `certificate_coefficient_identity` in the CertificateData section checks this equality by kernel reduction of exact rational arithmetic. The SymmetricCertificate section proves that collection, scaling, multiplication by V, and canonical sorting

preserve evaluation. The OrbitBridge and SquareSemantics sections connect those operations to (10) and (16). Consequently the finite coefficient check proves (17) for every dimension.

Now take $z_i = c_i^2$, so $z_i \geq 0$ and $V = 1$. The right side of (17) is nonnegative. By (14), the left side is $3/16 - \mathbb{E}P(\sqrt{3}S)$. This proves (6). Combining (5) with (6) proves (3).

6 Proof of the sharp Fourier bound

Let $W = W_1(f)$. If $W = 0$, inequality (1) is immediate. Otherwise put $c_i = b_i/\sqrt{W}$ and $S = \sum_i c_i X_i$. Then

$$\sum_i c_i^2 = 1, \qquad \mathbb{E}[f(X)S] = \frac{\sum_i b_i^2}{\sqrt{W}} = \sqrt{W}.$$

Set $Z = \sqrt{3}S$. Its distribution is symmetric, because complementing every sign replaces Z by $-Z$. Since $9/4 \geq 0$, pointwise

$$(Z - 9/4)_+ + (-Z - 9/4)_+ = (|Z| - 9/4)_+.$$

Therefore (3) implies

$$\mathbb{E}(Z - 9/4)_+ = \frac{1}{2}\mathbb{E}(|Z| - 9/4)_+ \leq 3/32. \tag{18}$$

For a Boolean-valued f, the elementary pointwise inequality

$$fZ \leq \frac{9}{4}f + (Z - 9/4)_+ \tag{19}$$

holds in both cases $f = 0$ and $f = 1$. Taking expectations, using $\mathbb{E}f = 1/8$ and (18), gives

$$\sqrt{3}\sqrt{W} = \mathbb{E}[fZ] \leq \frac{9}{32} + \frac{3}{32} = \frac{3}{8}.$$

Both sides are nonnegative, so squaring yields $3W \leq 9/64$, which is (1).

For the subcube indicator $f = \mathbf{1}\{X_1 = X_2 = X_3 = 1\}$, the first three coefficients b_i are $1/8$, and all remaining coefficients vanish by independence and zero mean of the unused sign. Hence $W_1(f) = 3/64$, proving attainment in every $n \geq 3$.

7 Average distance

Let A have density $a > 0$ and $f = \mathbf{1}_A$. For a uniform point of A, its ith coordinate mean is b_i/a. If U, V are independent uniform points of A, then

$$\mathbb{E}d_H(U, V) = \frac{1}{2}\sum_i (1 - \mathbb{E}[U_i V_i]) = \frac{n}{2} - \frac{W_1(f)}{2a^2}. \tag{20}$$

At $a = 1/8$, substitute (1) to get (2). For a codimension-three subcube, (20) and its exact Fourier weight give equality. The Lean development defines the weighted average as the double uniform-cube average of $f(x)f(y)d_H(x, y)$, divided by $(\mathbb{E}f)^2$; for an indicator this is exactly the ordered-pair average over A.

8 Formalization and exact certificates

The companion formalization uses Lean 4.19.0 and a pinned mathlib revision. It includes the finite rational identities and their interpretation for every cube dimension. Polynomial identities use `ring`. Finite rational table equalities use `decide +kernel`. The moment recurrence then connects these identities to the uniform probability measure on the Boolean cube.

A separate Python checker uses exact rational arithmetic. It checks all thirty symmetric coefficients, all twenty-eight positive weights, sixteen Bernstein intervals, both polynomial factorizations, both tail expansions, interval coverage, and the contact values. The reproduction package gives the commands and dependency versions. These checks use the stated certificate. They do not depend on a numerical optimization solver.

A Exact polynomial

In $P(t) = \sum_{j=0}^{9} p_j t^{2j}$, use:

j	p_j
0	$\frac{2065204456063070436648471 09}{362500000000000000000000000000}$
1	$\frac{-195397256376749992156612335 1}{217500000000000000000000000000000}$
2	$\frac{2814044752324915893803393 47}{489375000000000000000000000000000}$
3	$\frac{7409017415686803346331558 09}{244687500000000000000000000000000}$
4	$\frac{-3247861969010612115496283 47}{978750000000000000000000000000000}$
5	$\frac{796521088493109}{50000000000000000000}$
6	$\frac{-41161541352553493}{100000000000000000000000}$
7	$\frac{29617055737416117}{500000000000000000000000000}$
8	$\frac{-5574570362723033}{125000000000000000000000000}$
9	$\frac{341703734117989}{250000000000000000000000000}$

B Exact nonnegative symmetric identity

Use (17) with the following terms. The monomial weights below are in the **ordered-orbit** convention of (10), matching the CertificateData section. In `exact_certificate.json`, monomial entries use the conventional m_λ normalization; dividing those five weights by $\mathrm{aut}(\lambda)$ gives the weights printed here. The square terms use the same ordered-injection convention in both representations.

Every numerator and denominator in the final column is positive. This table and Appendix A completely determine identity (17).

r	H_r	a	b	w_r
1	$O_{(3,1,1,1,1,1,1)}$	–	–	$\frac{16646137361744047018987645 87}{67968750000000000000000000000}$
2	$O_{(2,2,2,2,1)}$	–	–	$\frac{10898934482037153838833003 1}{45312500000000000000000000}$
3	$O_{(2,2,2,1,1,1)}$	–	–	$\frac{36368430041679577673356417 9}{11328125000000000000000000}$
4	$O_{(2,2,1,1,1,1,1)}$	–	–	$\frac{23369443801983910767596555 27}{45312500000000000000000000}$
5	$O_{(2,1,1,1,1,1,1,1)}$	–	–	$\frac{21678902901545427631303627 3}{67968750000000000000000000}$
6	$G_{(7);a,b}$	(3)	-1	$\frac{49317944388734216342783360 3}{13593750000000000000000000}$
7	$G_{(6,1);a,b}$	$(4,-1)$	-1	$\frac{81276582295211350007274538 9}{76464843750000000000000000}$
8	$G_{(5,2);a,b}$	$(0,3)$	-1	$\frac{62167895823868919398383790 542583}{70837031250000000000000000000000}$
9	$G_{(5,2);a,b}$	$(1,2)$	-1	$\frac{16349742672694910124756998 3295373}{23612343750000000000000000000000}$
10	$G_{(5,2);a,b}$	$(3,0)$	-1	$\frac{40898714423911742716816466 080817}{70837031250000000000000000000000}$
11	$G_{(5,1,1);a,b}$	$(0,1,-4)$	1	$\frac{31631429016033966500059901 3407}{24468750000000000000000000000000}$
12	$G_{(5,1,1);a,b}$	$(1,-4,0)$	1	$\frac{56165887401691366342797622 43}{30585937500000000000000000000000}$
13	$G_{(5,1,1);a,b}$	$(2,-2,3)$	-1	$\frac{11461831387776513756474092 1259}{24468750000000000000000000000000}$
14	$G_{(4,3);a,b}$	$(1,-1)$	0	$\frac{53391677322784020922289982 06713}{65589843750000000000000000000000}$
15	$G_{(4,2,1);a,b}$	$(1,-4,-3)$	2	$\frac{31073909255556627765856054 8518749}{70837031250000000000000000000000}$
16	$G_{(4,2,1);a,b}$	$(1,-3,-1)$	1	$\frac{39203457027911089552018539 0303}{34981250000000000000000000000000}$
17	$G_{(4,2,1);a,b}$	$(1,2,0)$	-1	$\frac{12299856119437862451844941 3573949}{14167406250000000000000000000000000}$
18	$G_{(4,2,1);a,b}$	$(3,-1,4)$	-2	$\frac{30884554274001200121651411 444769}{47224687500000000000000000000000}$
19	$G_{(4,2,1);a,b}$	$(4,-1,3)$	-2	$\frac{15321637693779276596484703 530539}{29515429687500000000000000000000}$
20	$G_{(3,3,1);a,b}$	$(0,3,0)$	-1	$\frac{19163177340430193809351035 52589}{98384765625000000000000000000}$
21	$G_{(3,3,1);a,b}$	$(1,-3,-1)$	1	$\frac{23562437161184671324402035 11597}{19676953125000000000000000000}$

r	H_r	a	b	w_r
22	$G_{(3,2,2);a,b}$	$(1,-3,-1)$	1	$\frac{42477332960852370135610213284689}{20988750000000000000000000000000}$
23	$G_{(3,2,2);a,b}$	$(1,1,4)$	-2	$\frac{123346068448463558503725606137}{67851562500000000000000000000}$
24	$G_{(3,2,2);a,b}$	$(3,-1,1)$	-1	$\frac{346574630387532893541592905802 69}{62966250000000000000000000000000}$
25	$G_{(3,2,1,1);a,b}$	$(0,1,-3,0)$	0	$\frac{11186834185433279625736749 4399}{13593750000000000000000000000}$
26	$G_{(2,2,2,1);a,b}$	$(1,0,0,-3)$	0	$\frac{12902157725546495394762981 9}{45312500000000000000000000}$
27	$G_{(2,1,1,1,1,1);a,b}$	$(1,-4,0,0,0,0)$	0	$\frac{22348383362905355092020478147}{67968750000000000000000000000}$
28	$G_{(2,1,1,1,1,1);a,b}$	$(1,-3,0,0,0,0)$	0	$\frac{20937950969712400953131202529}{33984375000000000000000000000}$

Research and publication

This work is fully autonomous agentic research generated by AI systems. AI agents produced the mathematical arguments, formal proofs, computations, and manuscript. Dev D. Goyal is responsible for publication.

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